Urban Heat Mapping with Agentic AI for Sustainable Cities

Theme: Sustainable Cities and Communities — Addressing SDG 11 by detecting urban heat hotspots and recommending targeted cooling interventions.

Problem Statement: Urban Heat Island (UHI) effects in developing cities like Faisalabad increase health risks, energy consumption, and environmental stress. Without data-driven mapping, it’s hard for urban planners to know where to focus cooling solutions like green spaces, rooftop gardens, or reflective materials.

Proposed Solution: We will use agentic AI with IBM watsonx and Google Earth Engine to automatically collect satellite-derived Land Surface Temperature (LST) data for a target city, detect and visualize urban heat hotspots, and generate actionable recommendations for cooling interventions. The workflow is automated so it can be adapted to other cities with minimal manual work.

Approach & Workflow:

1. Data Acquisition: Google Earth Engine, Landsat 8 Collection 2 Level-2 Surface Temperature (ST_B10), AOI from geojson.io for Faisalabad, Date range May–June 2024.

2. Preprocessing: Cloud filtering (CLOUD_COVER < 10%), median composite creation, LST conversion from Kelvin to Celsius.

3. Analysis: Identify hotspots above 38°C, classify into Cool (<30°C), Moderate (30–35°C), Hot (>35°C), Extreme (>40°C).

4. Visualization: Interactive heatmap using Folium, export GeoTIFF to Google Drive.

5. Agentic AI Integration: IBM watsonx.ai to summarize findings and recommend cooling strategies.

Technologies & Tools:

Google Earth Engine	Remote sensing data processing
Landsat 8 Level-2	Surface Temperature measurement
Python (Colab)	Automation & scripting
Folium	Interactive map visualization
GeoJSON.io	AOI selection
Google Drive	Data export & storage
IBM watsonx.ai	AI insights & recommendations
Git/GitHub	Version control

Expected Outputs:

- GeoTIFF raster of LST for Faisalabad in °C.
- Interactive web map of heat hotspots.
- AI-generated PDF report with findings and recommendations.

Impact & Scalability: This approach can be run for any city with free satellite data, helping city planners prioritize cooling investments and scale to regional/national levels without additional data costs.

Glossary:

AOI (Area of Interest): The region you want to analyze.
Bounding Box: A rectangular selection around an AOI.
GeoJSON: A text-based format to describe geographic shapes.
LST (Land Surface Temperature): How hot the ground's surface is.
Cloud Masking: Removing cloudy pixels to get a clearer image.
GeoTIFF: A georeferenced image file usable in GIS software.
Agentic AI: AI agents that reason, plan, and act on data to produce meaningful outcomes.